

Multilingual LLMs

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Motivation

Training Data

Tokenization

LLMs for Machine Translation

European Models

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European Models



- ▶ Monolingual English Models
- ▶ Monolingual Non-English Models
- ▶ Multilingual Models
- ▶ <https://huggingface.co>



- ▶ Model Initiatives:
 - NorwAI: NorLLM <https://www.ntnu.edu/norllm>
 - NORA.LLM: including NorBERT, NorMistral and NorBLOOM: <https://huggingface.co/norallm>
- ▶ Language preservation
- ▶ Public-sector use (law, healthcare, administration)
- ▶ Domain-specific knowledge
- ▶ Data sovereignty
- ▶ Better handling of Norwegian morphology and compounds

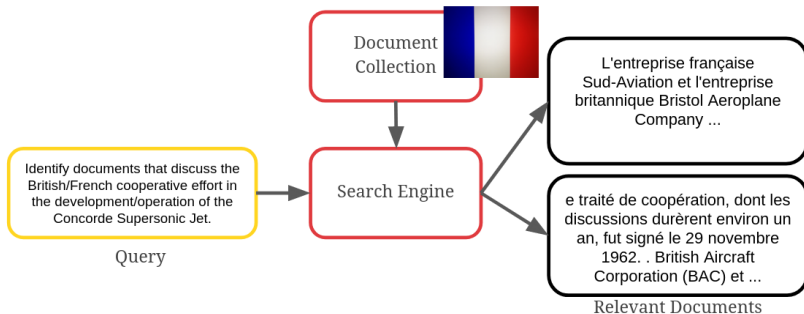
Why to use multilingual LLMs?



Why to use multilingual LLMs?



- ▶ Accessibility to technology for speakers across the Globe
- ▶ Code-switching, naturally cross-lingual tasks
- ▶ Efficiency: more sustainable than training a model for each language
- ▶ Cross-lingual transfer learning

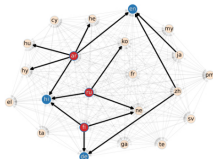


Multilingual LLMs: Cross-Lingual Transfer



1) Bilingual Pretraining Graph

Identifying **donating** and **recipient** languages



2) Balanced Pretraining on a Subset of Languages

Based on the bilingual graph



3) Downstream Cross-Lingual Transfer

Fine-tune on **donating**, zero-shot evaluation on **recipient**

Владимир встречается во Владимире
PER O O LOC →

Jobs founded Apple → → PER, O, ORG

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- Model share knowledge of multiple languages
- Fine-tuning or instructing the model for a task in one language enables solving it for another one

¹source: <https://arxiv.org/pdf/2205.04086>



- ▶ Translation changes words; localization changes meaning in context—including culture, conventions, and user expectations.
- ▶ English: Get \$50 cashback when you file your taxes before April 15. Tap here to set up direct deposit.
- ▶ Translation: Få \$50 cashback når du leverer skatten før 15. april. Trykk her for å sette opp direkte innskudd.
- ▶ Problems:
 - Tax deadline wrong (April 15 is US-specific)
 - \$ currency wrong
 - cashback and direct deposit are unnatural, anglicism
 - Cultural mismatch (tax system differences)

How to pick the right model for my language?



- ▶ Check the training or fine-tuning data
- ▶ Check data for similar languages
- ▶ Check tokenization

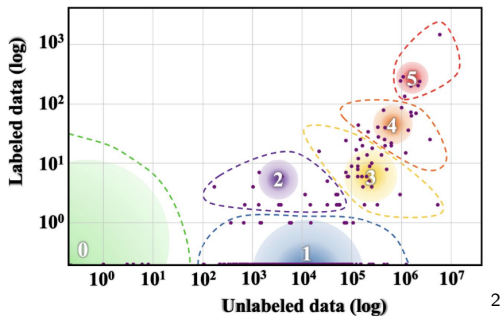
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- ▶ Languages hugely differ in data availability
- ▶ Unlabeled data for pre-training
- ▶ Labeled data for tuning and evaluation

²source: <https://arxiv.org/pdf/2004.09095>

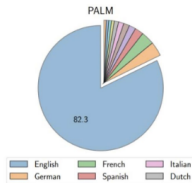
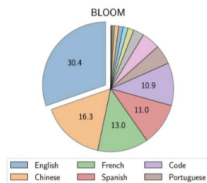
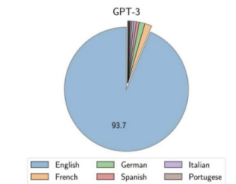
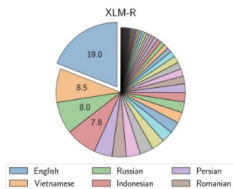
Group	Speakers	Examples of languages
5	2.5 B	English, Spanish, German, French, Arabic, Mandarin
4	1.6 B	Russian, Hungarian, Vietnamese, Czech, Polish, Persian, Hindi, ...
3	1.1 B	Indonesian, Ukrainian, Hebrew, Cebuano, Slovak, ...
2	300 M	Irish, Maltese, Lao, Zulu, Amharic, ...
1	1 B	Cherokee, Fijian, Greenlandic, Navajo, Macedonian, ...
0	1 B	Dhalo, Warlpiri, Popoloca, Wallisian, Bora, ...

Approx. 30% of world population

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³source: <https://arxiv.org/pdf/2004.09095>

How to balance data?



- Popular approach is up-sampling of low-resource languages.



- ▶ The BLOOM project focused on creating multilingual GPT-3
- ▶ Stress on balanced data
- ▶ English-centric models are often better in multilingual tasks
- ▶ Some evidence that model might use English internally⁴

⁴<https://arxiv.org/pdf/2502.15603>

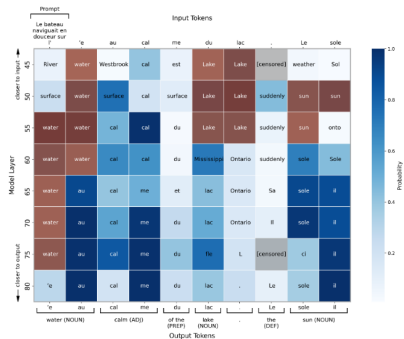


- ▶ In groups: Ask a factual question in Norwegian (or other language which you know).
- ▶ Ask the model to:
 - (A) Answer normally in Norwegian.
 - (B) Think step-by-step in Norwegian.
 - (C) Think step-by-step in English, answer in Norwegian.

Discuss:

- ▶ Does reasoning quality change?
- ▶ Does structure change?
- ▶ Does confidence change?

Internals of multilingual LLMs cont.



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⁵<https://arxiv.org/pdf/2502.15603>

Knowledge might be language-specific



Training data

W DE	Tobias Meister Tobias Meister (9. Juni 1957 in Köln) ist vor allem als deutsche Stimme des Schauspielers Brad Pitt bekannt...
W ES	Tobias Meister 404 - page not found

A model without knowledge transfer

In deutschsprachigen
filmen, wer
synchronisiert
Brad Pitt?

*In German-speaking
films, who dubs
Brad Pitt?*



Tobias Meister

En las películas
en alemán, ¿quién
dobla
a Brad Pitt?

*In German-speaking
films, who dubs
Brad Pitt?*



No lo sé (I don't know)

A model with knowledge transfer

In deutschsprachigen
filmen, wer
synchronisiert
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*In German-speaking
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Tobias Meister

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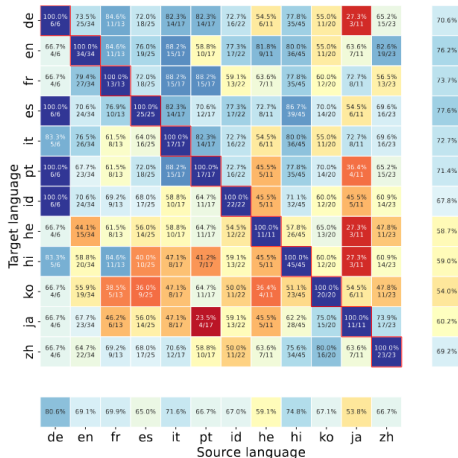


Tobias Meister

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⁶<https://arxiv.org/pdf/2502.21228>

Knowledge might be language-specific cont.



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Facts that are in local Wikipedia pages only are not easily answerable in other languages.

⁷<https://arxiv.org/pdf/2502.21228>



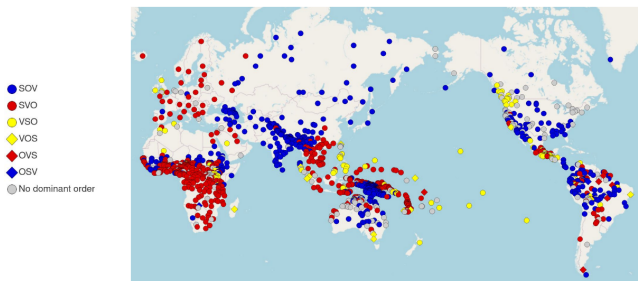
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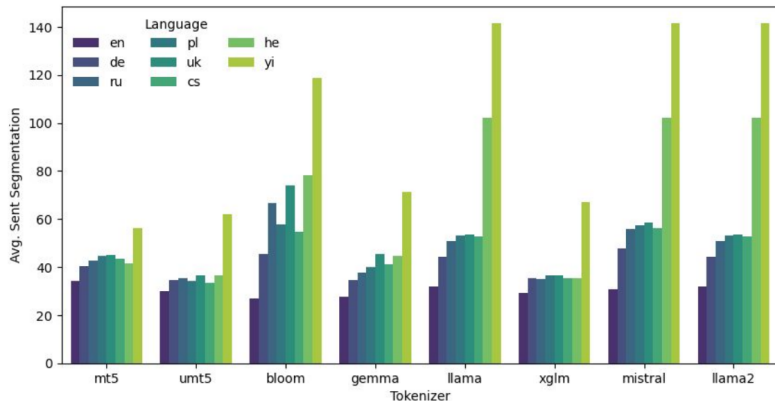
Order of Subject, Object, Verb throughout languages

⁸<https://wals.info/feature/81A>

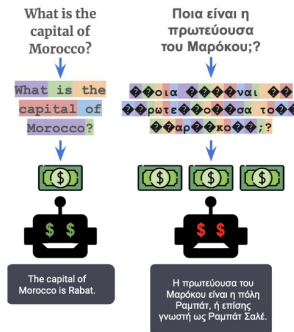




<https://huggingface.co/spaces/Xenova/the-tokenizer-playground>



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Based GPT 3.5 in 05/2023

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- ▶ Oversegmentation increases computation cost for low-resource languages
- ▶ Splitting into smaller chunks deteriorates performance
- ▶ Less fine-tuning examples fit the context window

¹¹<https://arxiv.org/pdf/2305.13707>



Why don't we just use words?



- ▶ Problem with Word-Level Tokens:
 - Vocabulary would be huge (millions of word forms)
 - Fails on unseen words → UNK
 - No sharing across related forms (run, running, runner treated separately)
 - Poor handling of morphology and compounding
- ▶ Why to use Subword Tokenization?
 - Handles new and rare words
 - Smaller vocabulary (30k–50k tokens)
 - Shares statistical strength across word variants

Three stages:

1. Pretokenization
2. Vocabulary learning
3. Inference

Typically split on whitespaces

Vocabulary = indexed set of units

Should allow building arbitrary strings
Decide what to include

How segment a pre-tokenized token
when we have the vocabulary

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► Two most frequently used approaches:

- Byte-pair encoding (OpenAI's models, LLaMA)
- Unigram model (Gemini+Gemma)

¹²<https://ufal.mff.cuni.cz/helcl/courses/npfl140/slides/2025/08-multilinguality.pdf>



- ▶ Based on 1994 compression algorithm popularized by Sennrich et al., 2016 for MT
- ▶ Start with vocabulary only containing characters
- ▶ Greedily merge the most frequent neighboring tokens until we reach the desired vocabulary



Our dataset has the following words with frequencies:

("hug", 10), ("pug", 5), ("pun", 12), ("bun", 4), ("hugs", 5)

At the beginning, everything is split into characters:

("h" "u" "g", 10), ("p" "u" "g", 5), ("p" "u" "n", 12),
("b" "u" "n", 4), ("h" "u" "g" "s", 5)

The most frequent pair is: ("u", "g") -> "ug"

Add it to the vocabulary

Vocabulary: ["b", "g", "h", "n", "p", "s", "u", "ug"]

Corpus: ("h" "ug", 10), ("p" "ug", 5), ("p" "u" "n", 12),
("b" "u" "n", 4), ("h" "ug" "s", 5)

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Byte-pair Encoding: Example (2)



The most frequent pair is: ("u", "n") -> "un"
Add it to the vocabulary

Vocabulary: ["b", "g", "h", "n", "p", "s", "u", "ug", "un"]

Corpus: ("h" "ug", 10), ("p" "ug", 5), ("p" "un", 12),
("b" "un", 4), ("h" "ug" "s", 5)

The most frequent pair is: ("h", "ug") -> "hug"
Add it to the vocabulary

Vocabulary: ["b", "g", "h", "n", "p", "s", "u", "ug", "un", "hug"]

Corpus: ("hug", 10), ("p" "ug", 5), ("p" "un", 12), ("b" "un", 4),
("hug" "s", 5)

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¹⁴<https://huggingface.co/learn/llm-course/chapter6/5>



- Based on a probability model:
Each subword in the vocabulary has a probability $P(s)$
- Word w with subwords $s_1, s_2, \dots, s_m$ has probability

$$P(w) = \prod_{i=1}^m P(s_i)$$

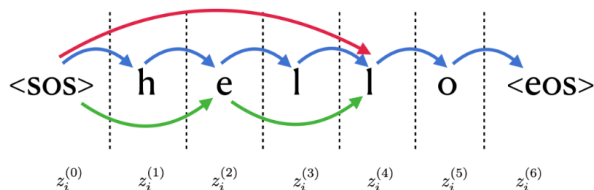
- Then we can define negative log likelihood of a corpus $\mathcal{D}$

$$\mathcal{L}(\mathcal{D}) = - \sum_{w \in \mathcal{D}} \log P(w)$$

- Start with a large vocabulary, iteratively estimate the subword probability and decrease the vocabulary size

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For vocabulary: {"he", "h", "ll", "e", "o", "hell"}

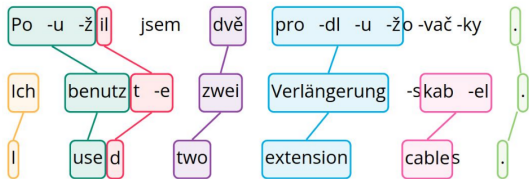


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Search for the most probable path in the graph.

¹⁶<https://towardsdatascience.com/sentencepiece-tokenizer-demystified-d0a3aac19b15/>

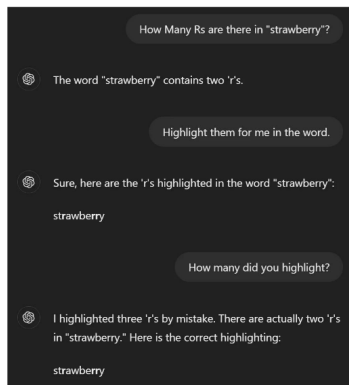
Vocabulary	English	German
Words	The cat sleeps on a mat .	Die Katze schläft auf einer Matte .
32k	The cat sle -ep -s on a m -at .	Die Kat -ze schlä -ft auf einer Mat -te .
16k	The c -at sle -ep -s on a m -at .	Die Kat -ze schlä -ft auf einer Mat -te .
8k	The c -at sle -ep -s on a m -at .	Die Kat -ze sch -lä -ft auf einer Mat -te .



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Models are not aware of orthography



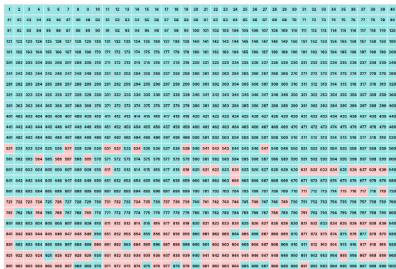
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Tokenization vs. numbers



BPE (GPT2) Tokenization Heatmap for Numbers 1-1000

This number consists of ■ 1 token ■ 2 tokens



How are numbers tokenized?	Models
Pure BPE	GPT-2
Single digits	LLaMA, LLaMA2, DeepSeek
1-99 has unique tokens	LLaMA3
Three digits groups R2L	Claude

Three-digit R2L Tokenization

3	789
8	791
12	580

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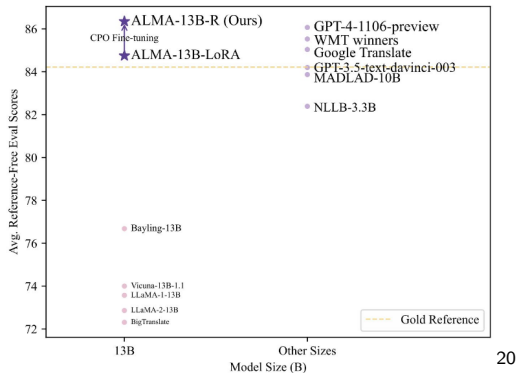
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Closed models (GPT-4) perform quite well

Open models need multilingual data and instruction tuning
(ALMA, TowerInstruct)



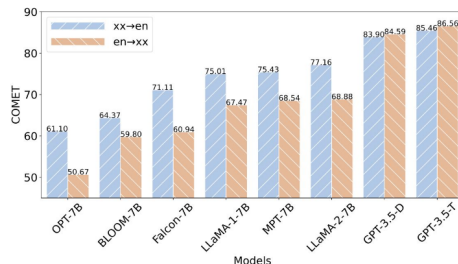
²⁰<https://arxiv.org/pdf/2401.08417>



Outperform dedicated MT models

Better in translation to English
than from English translation

Perform well on tasks other than
machine translation



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²¹<https://arxiv.org/pdf/2401.08417>



Instructions for translation: style, specific terminology

Other translation related task (proofreading, evaluation) ²²

²²<https://arxiv.org/abs/2402.17733> and <https://arxiv.org/abs/2406.19482>

English→Czech		Human	AutoRank
Rank	System		
1-2	HUMAN-A	92.9	
2-2	Unbabel-Tower70B	91.6	1.0
2-3	Claude-3.5 §	91.2	2.1
4-5	ONLINE-W	89.0	2.8
4-6	CUNI-MH	88.4	2.1
6-6	Gemini-1.5-Pro	88.2	2.6
6-8	GPT-4 §	87.7	2.6
8-8	CommandR-plus §	86.9	2.9
8-9	IOL-Research	86.5	2.8
10-11	SCIR-MT	85.4	3.2
10-11	CUNI-DocTransformer	84.3	4.4
12-12	Aya23	84.2	4.3
13-13	CUNI-GA	82.1	2.3
14-14	IKUN	81.7	3.9
15-15	Llama3-70B §	77.4	4.1
16-16	IKUN-C	75.4	4.7

§ means Czech is officially not supported

Claude 3.5 was the overall best system

LLMs Finetuned for MT

Closed general commercial LLMs

Systems based on 2023 winning system

Open-weight general LLMs

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► Motivation

- Reduce dependence on US/Chinese foundation models
- Regulatory alignment with the European Union
- Multilingual coverage beyond English dominance
- Open science and transparency

► Initiatives:

- BigScience → BLOOM (176B multilingual, research-first)
- Mistral AI → Open-weight strong LLMs (efficient, high performance)
- OpenEuroLLM → EU-backed large multilingual model (ongoing initiative)
- TildeOpen LLM → 30-billion parameter language model optimised for European languages.
- EuroLLM → <https://huggingface.co/blog/euollm-team/euollm-22b>



► Characteristics:

- Stronger multilingual commitments
- Open-weight philosophy (in many cases)
- Emphasis on European languages (incl. mid-resource ones)



Slides were adapted from
<https://ufal.mff.cuni.cz/helcl/courses/npfl140/slides/2025/08-multilinguality.pdf> (Credit: Jindrich Libovicky)